

Supplementary Information

This document is intended as an archive of relevant information that were not immediately pertinent to the main report. For raw data and code generated and used in the project, refer to our project wiki.

Error-prone PCR

Error-prone PCR was used not as a method for generating variants for directed evolution, but as a benchmarking process so that the tag variants generated can be compared to those generated by the computational pipeline. One particular difficulty encountered during our error-prone PCR process was samples with low concentration that showed up as smears on gel electrophoresis imaging. Therefore, we carried out several troubleshooting experiments to encounter these, ranging from optimising Manganese and Magnesium ion concentrations, to changing the ratios of the primers. According to existing literature, it was found that changing the concentration of Mn^{2+} ions in the range of 0-0.5 mM affected the frequency of mutant generation in epPCR ([Le et al. 2013](#)). Furthermore, Taq polymerase was chosen over Pfu as it has a lower intrinsic 3' to 5' exonuclease activity and therefore a higher rate of mutations ([Cline et al. 1996](#)). Additionally, existing literature has also shown that primer concentrations can be optimised in PCR reactions to optimise yield by simultaneously changing both forward and reverse primer concentrations ([Green et al. 2018](#)).

Selection

During the liquid batch selection process, chemically transformed bacterial DH10B cells containing pBAD, B-lactamase or B-lactamase plus SSB, NEXT or De Novo were induced with 20% L-arabinose solution and grown in carbenicillin media, with carbenicillin containing growth media concentrations ranging from 0, 5×10^3 , 10×10^3 , 30×10^3 , 40×10^3 ug/ml carbenicillin concentrations. Our hypothesis was that IDP tag improves the solubility of B-lactamase protein, therefore improving resistance to carbenicillin and therefore cells with tags show higher survival rate. Our results show that pBAD plasmid plus B-lactamase alone had a lower survival rate than cells that contain a tag, particularly at very high carbenicillin concentrations. Furthermore, some of the results also show that the degenerate evolved counterpart of our tags have higher survival rate in high concentrations of carbenicillin than the original variant of the tag, indicating successful directed evolution.

Initially, the aim was to carry out a functional enzymatic activity assay to ensure that our IDP tag does not significantly affect target protein activity, as well as a solubility assay to investigate the effect of tag on solubility.

Viability of Carbenicillin Selection System

In silico generation of variants and the compression into degenerate sequences led to a potential population of unknown variant performance. The selection system is shown to be able to use carbenicillin concentration as a proxy for population fitness. OD data indicates trends we expect: the death of low performing variants corresponding to lower OD; the higher concentration of carbenicillin causing slower population recovery, and the final incubation indicating tag performance.

Based on pBAD and pBAD/beta lactamase controls, the selection system works for differentiating variants using OD as proxy for population and carbenicillin as selection pressure.

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OD data indicated that pBAD controls completely died off in the presence of carbenicillin whereas pBAD/beta lactamase control displays resistance up to 15k ug/mL. This proved that the selection system works for differentiating variants using OD as proxy for population and carbenicillin acts as a selection pressure.

The addition of carbenicillin to cell cultures produced concentration associated differences in the endpoints OD600 readings of SSB and NEXT derived libraries. The increase in carbenicillin concentration decreased the OD readings of each strain, with the tagged beta-lactamase displaying lower sensitivity to carbenicillin. This rescue of the beta-lactamase at 30-40k ug/mL indicates the presence of a variant that successfully rescues the beta-lactamase.

Viability of In Silico Generation & Scoring of Tag

In silico selection of generated mutants showed hotspots in the sequence logo which reappear in alternative random seeds indicating significant likelihood that selection is highlighting regions of high importance in optimizing parameters for EB mechanism.

Target protein selection

We searched protein databases such as UniProt to shortlist a list of target proteins and select a final of 3. We categorised our targets into three main groups, one protein that we could use to measure enzymatic activity and ensure our tags don't negatively affect it, one that has been more difficult to improve in solubility with traditional tags like MB, and one that has been shown with strong literature evidence to improve in solubility with many tags. On top of that, we also look for interesting properties of these proteins, meaning that increasing the solubility of these proteins should optimally help future therapeutic or industrial research. Our finalised list of target proteins included proteins suitable in terms of small size and complexity, such as TEV protease ([Yu et al. 2017](#)), MMP13 ([Marblestone et al. 2006](#)), IL3 ([Nguyen et al. 2024](#)), taCA, isPETase, hEGF and GFP ([Jo, 2022](#)), and mTDT ([Tang et al. 2024](#)). Of these the three proteins taCA, isPETase and hEGF were selected for our final validation round. This is because we aim to investigate enzymatic assays with isPETase and taCA, and hEGF was small enough to be ordered as a ready plasmid. Additionally, we also chose two commercially available and tested tags for benchmarking comparisons with our own evolutionary derived tags. These were MBP and GB1 ([Zhou and Wagner, 2010](#)), which like our tags, both are able to be attached to C-terminus of target proteins, allowing direct and less biased comparison.

Gibson assembly of CAHS1

Two batches of Gibson assembly were performed under different conditions to insert the CAHS1 gene into the pBAD vector, using ssDNA and dsDNA version of the CAHS1 gene fragment. The reaction conditions are as follows:

In the ssDNA version, 3 pieces of DNA fragments were used for assembly: ssDNA CAHS fragment 1, dsDNA CAHS fragment 2, and dsDNA linearized pBAD plasmid backbone. 10 µL mixture of the 3 fragments were made with molar ratio between the 3 fragments being 200:5:1 respectively.

Subsequently, 10 µL of NEBuilder® HiFi DNA Assembly Master Mix was added to the mixture, and the mixture was incubated at 50 °C for 60 minutes.

In the dsDNA version, 3 pieces of DNA fragments were used for assembly: dsDNA CAHS fragment 1, dsDNA CAHS fragment 2, and dsDNA linearized pBAD plasmid backbone. Mixture of dsDNA CAHS fragment 2 and dsDNA linearized pBAD plasmid backbone were made first, with molar ratio between the 2 fragments being 5:1. Subsequently, 10 µL of NEBuilder® HiFi DNA Assembly Master Mix was added to the mixture, and the mixture was incubated at 50 °C for 30 minutes. Then, dsDNA CAHS fragment 1 was added to the mixture, topping up reaction volume to 20 µL with the molar ratio between the 3 fragments being 5:5:1 respectively. Finally, the mixture was further incubated at 50 °C for 30 minutes.

The products from each gibson reaction was transformed into competent DH10β E. coli cells and streaked onto agar plates containing 0.2% L-arabinose. Transformed cells containing empty pBAD vectors were observed through the expression of sfGFP causing fluorescence.

Table 1. Results of Gibson assembly of CAHS tag into pBAD vector using ssDNA and dsDNA gene fragments. Both transformations revealed contamination from cells containing empty pBAD vectors.

Batch	Sequenced?	T0 colonies	pBAD contamination	Outcome
ssDNA	Yes (21 variants confirmed)	Mostly non-fluorescent (library)	Present but minor	Usable library; freeze-thaw cycles 0–6 incubated; cycles 0,2,3 to be sequenced
dsDNA	No	Almost entirely fluorescent (pBAD)	Dominant	Library unusable; repurposed for WT vs pBAD survival control (result shown below)

Dry run of freeze-thaw selection

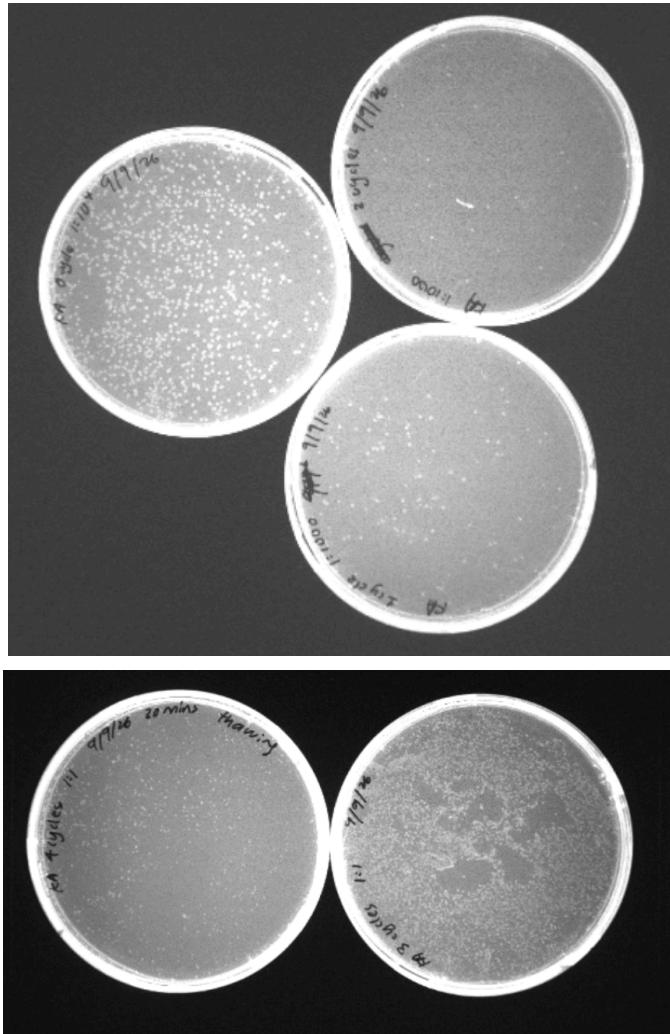


Fig. 1. Plate images from freeze-thaw dry run. The number of colonies were used to determine the dilution factor for cell plating.

The dilution series used for plating was calibrated based on a preliminary dry run performed with cells containing the empty pBAD vector only, on the assumption that the result would be comparable across constructs, as long as the OD was kept identical. This assumption proved wrong: the resulting dilution factor was too high when applied to this batch, yielding only 1 to 5 colonies per plate. Therefore, the resulting colony counts could not be seen as representative, and no usable data was acquired from this. The dilution factor was then corrected for the dsDNA Gibson Assembly used for subsequent freeze-thaw assays.

Samples from freeze-thaw cycles 0-6

Following overnight incubation, samples from flash freeze-thaw cycles 0-6 of the ssDNA Gibson Assembly products were streaked (rather than spread, due to culture saturation) onto plates containing L-arabinose.

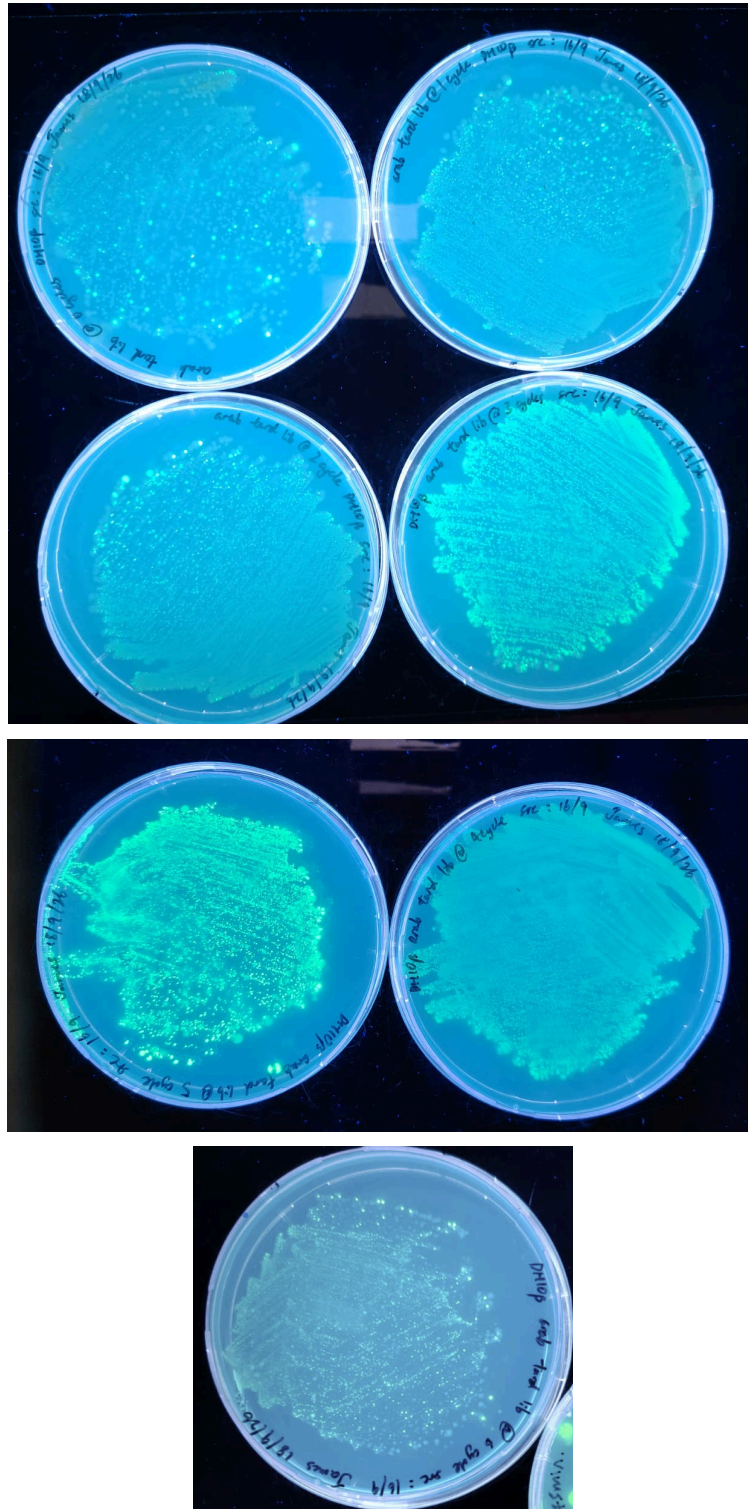


Fig. 2. Streaked plate images from flash freeze-thaw cycles 0-6.

As shown in the above figures, a clear trend was observed in the fluorescent colonies: plates from the earlier cycles (0-2) were predominantly non-fluorescent, indicating library dominance, whilst the proportion of fluorescent colonies (colonies containing the empty pBAD vector) increased substantially at later cycles, with fluorescence becoming the dominant phenotype by cycle 5.

We propose this shift reflects the differential growth kinetics between the library and pBAD populations during post-thaw outgrowth rather than differential freeze-thaw survival alone, as explained below.

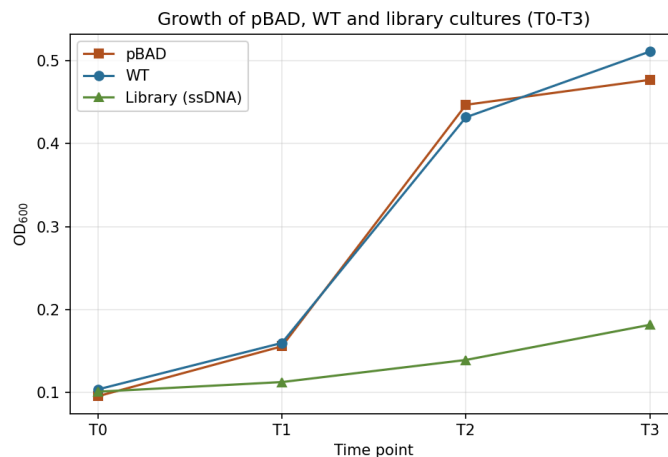


Fig. 3. Change in OD₆₀₀ of different cultures across different timepoints.

OD₆₀₀ (normalised to the standard 1 cm cuvette pathlength) was monitored at four time points for pBAD, wild type and library (ssDNA) cultures. Over the roughly 3-hour monitoring period, pBAD and WT cultures grew similarly, reaching an OD of 0.48 and 0.51 respectively, whereas the library culture grew markedly more slowly, reaching only 0.18 over the same interval.

Therefore, at low cycle numbers, the library's numeric majority in the starting culture is sufficient for it to remain dominant after overnight recovery despite its slower growth rate. However, repeated freeze-thaw cycles were able to progressively reduce the size of the surviving population, and recovering a much smaller founding population to saturation requires more rounds of doubling during outgrowth, which compounds pBAD's faster growth rate over more generations. We suggest that at higher cycle numbers, the library's numeric advantage is no longer sufficient to offset this effect, especially with being allowed to reach saturation with overnight incubation. This allows pBAD to increasingly dominate the recovered culture regardless of the underlying freeze-thaw survival ratio.

Nonetheless, this is presented as a hypothesis rather than a confirmed mechanism, as fluorescence proportion on streaked plates is qualitative and was not able to be quantified by colony counts.

Absence of empty-vector contamination in the WT culture

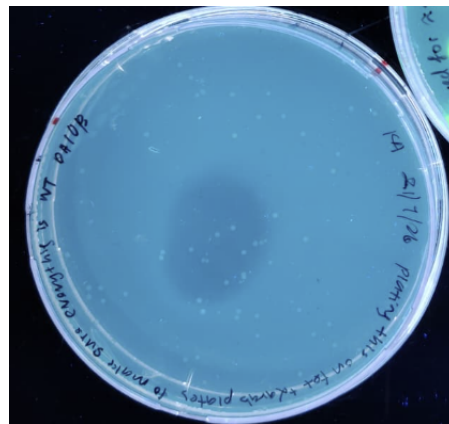


Fig. 4. Plate image of cells expressing wild-type CAHS1 after plating with arabinose.

To check that the WT culture was not contaminated with the empty pBAD vector, cells were plated with arabinose, under which empty-vector colonies express sfGFP and appear green. No green colonies were observed in the WT sample (Figure 4), indicating no detectable pBAD contamination.

In silico validation of tags

Wet lab results suggest that parameters used in in-silico selection are good proxies for protein performance. It is unknown whether the change in parameters from pre to post selection libraries are optimal. Further rounds of selection will inform whether parameters are optimal or whether the optimal is much further away. A refinement to the model is to find the optimal range by repeating selection rounds until parameters no longer fluctuate. This will then produce a model that can directly score the performance of each tag based on this range such that successful tags can be differentiated based on deviation from optimal parameter value.

SSB derived library comparison with original

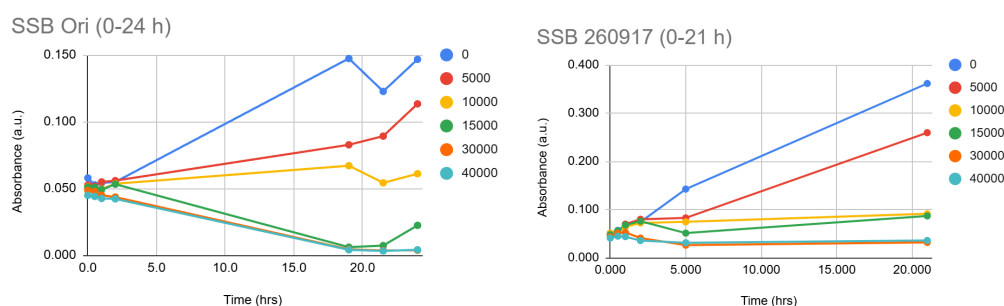


Fig. 5. OD changes of SSB original tag and library

Comparing the performance of the SSB derived library and the original tag, the library had generally lower performance over a period of 21 hours at all concentrations of carbenicillin, compared to the original tag where the OD600 readings varied significantly between different carbenicillin concentrations. This is within our expectations that large proportions of variants within the library have reduced solubility tag performance as compared to the original template.

NEXT derived library comparison with original

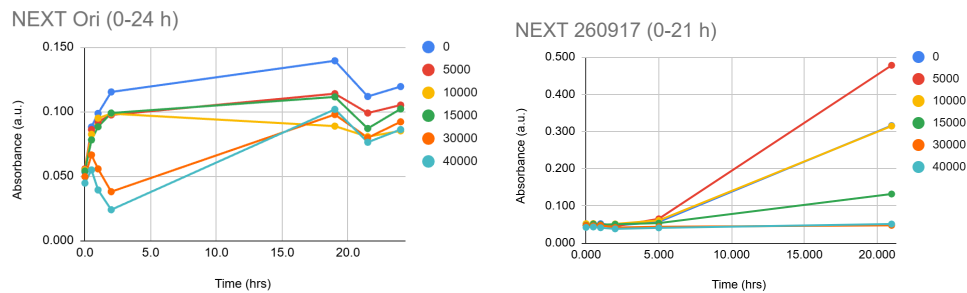


Fig. 6. OD changes of NEXT original tag and library

Original vs Library Performance NEXT

OD600 data shows that NEXT library is more sensitive to carbenicillin than the NEXT original. This could be due to the library containing a higher proportion of non-functional variants that fails to rescue the beta-lactamase. This would cause a lower initial population causing population recovery to be slow. To quantify the performance of the optimal selected tag after each selection round, it is suggested to repeat selection conditions on monoclonal populations.

Original vs Library Performance SSB

OD600 data shows that the SSB library is less sensitive to carbenicillin than SSB original. This is expected as SSB tag was not a known solubility enhancer and obtained through methodology described by (NEXT paper). This shows that our methodology is able to generate advantageous IDP variants that are better performing than non-evolved original templates.

Functional Assay

The beta-lactamase selection system partially validates non-disruptive impact on enzyme function as it was able to rescue the beta lactamase indicated by higher OD600 readings in 30k and 40k ug/mL concentrations.

Functional assays to quantify change in enzymatic activity were planned but failed to be carried out due to time constraints. taCA and isPETase were chosen targets due to the ability to act as an esterase. Enzyme activity was to be characterized through colorimetric assays to validate impact of the tag on the enzyme function.

Solubility Assay

Solubility assay was carried out via densitometric analysis. Due to time and budget constraints, the solubility assay was not carried out in completion and insufficient data to conclude.

CAHS core motif

In the main results section, only motif 1 of the full CAHS protein was analysed for specific relevant characteristics. The CAHS core including motif 1 and constant flanking regions was also analysed for population composition and cluster traits for the same properties applied to the NEXT and SSB

derived tags due to its intrinsically disordered nature. Since the CAHS core showed insignificant variations for NEXT and SSB specific IDP properties across selection, the results are not included in the main report.

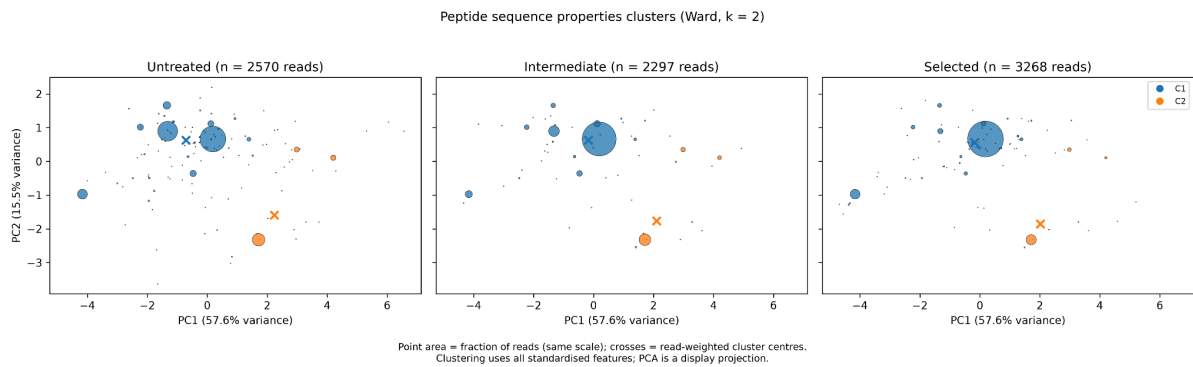


Fig. 7. Cluster diagrams of CAHS core at 0 (Untreated), 3 (Intermediate) and 6 (Selected) rounds of freeze-thaw selection.

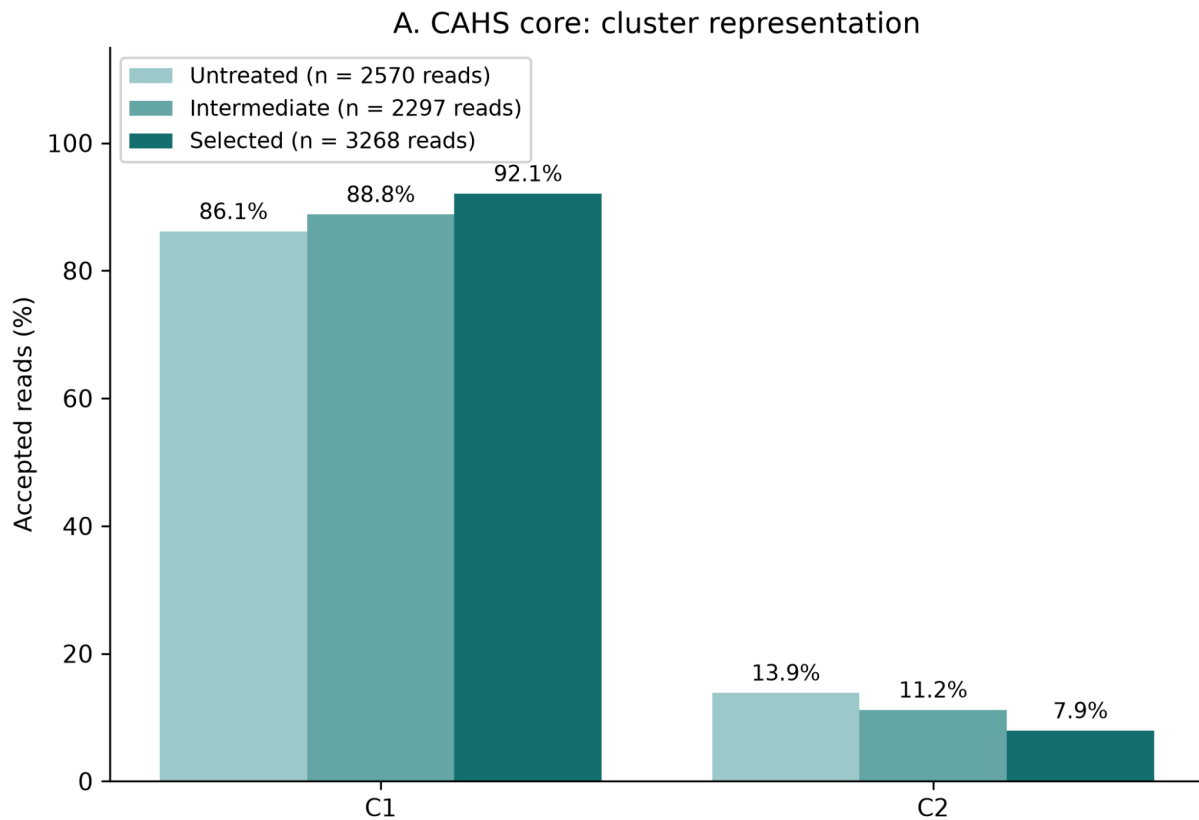
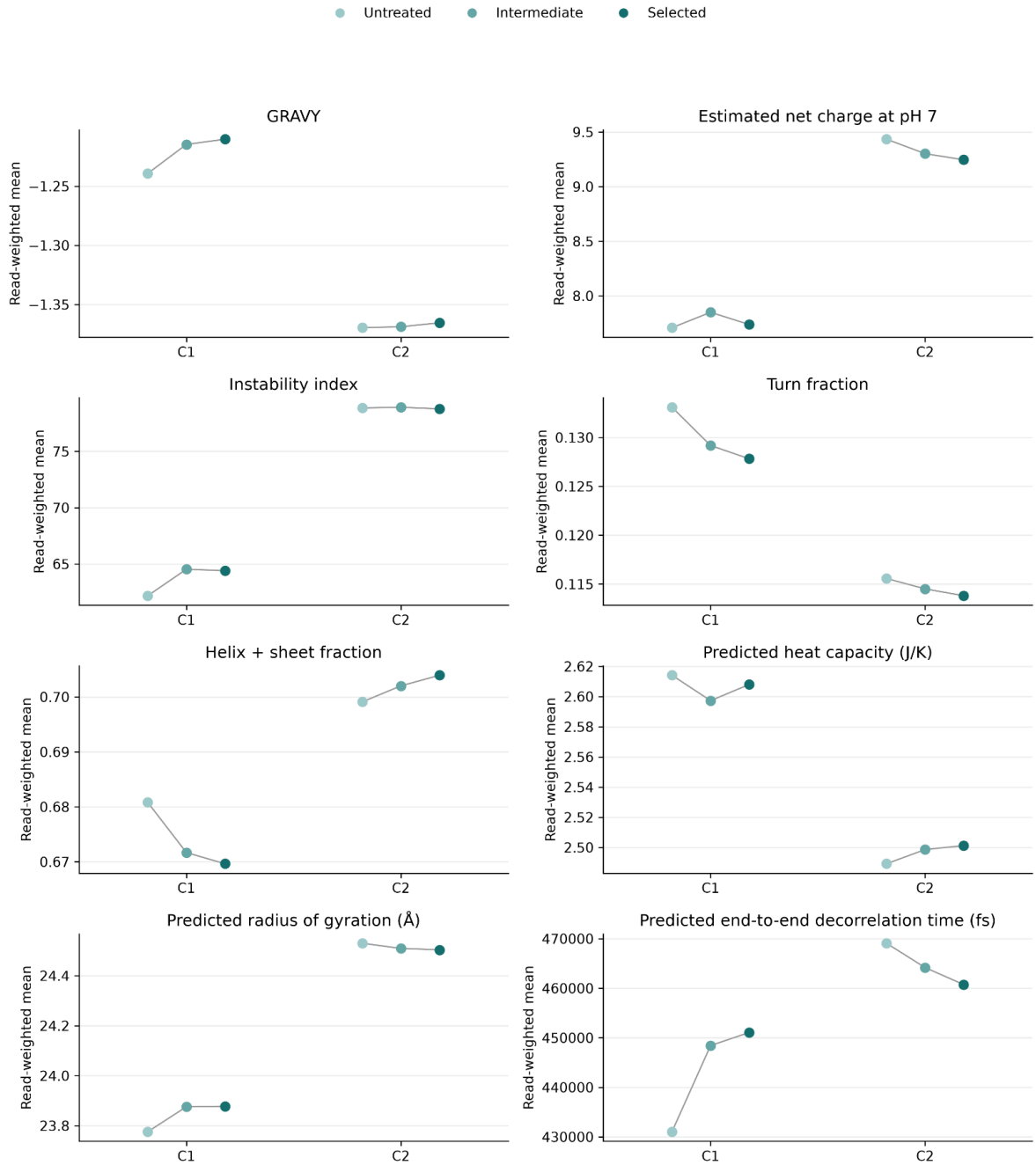


Fig. 8. CAHS core cluster representation at 0 (Untreated), 3 (Intermediate) and 6 (Selected) rounds of freeze-thaw selection.

B. CAHS core: traits within each fixed cluster



Each point is a within-cluster read-weighted mean; connecting lines show changes across samples.

Fig. 9. Read-weighted mean cluster properties at 0 (Untreated), 3 (Intermediate) and 6 (Selected) rounds of freeze-thaw selection.

Cluster representations for SSB and NEXT libraries

The cluster representations of each cluster for each tag are represented as below. The same data can be found in table format in the relevant sections under Results.

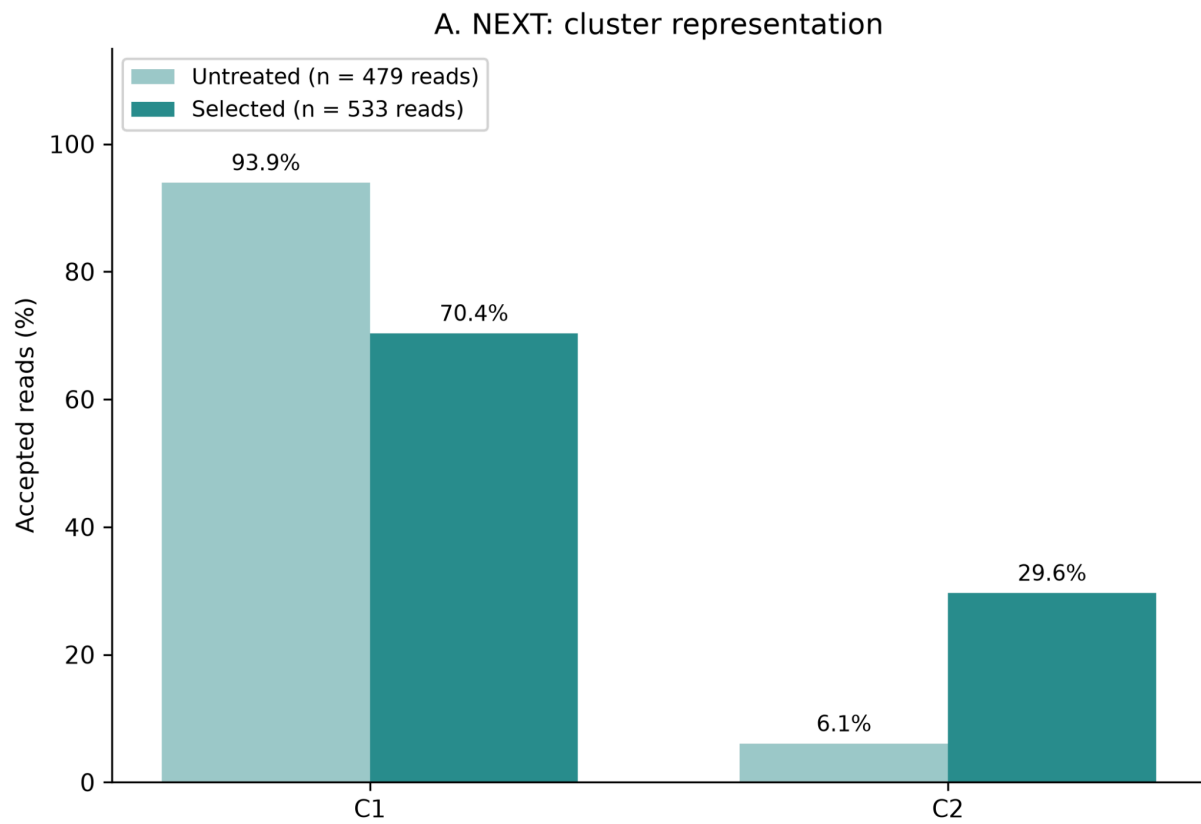


Fig. 10. Cluster representation of NEXT clusters before and after selection.

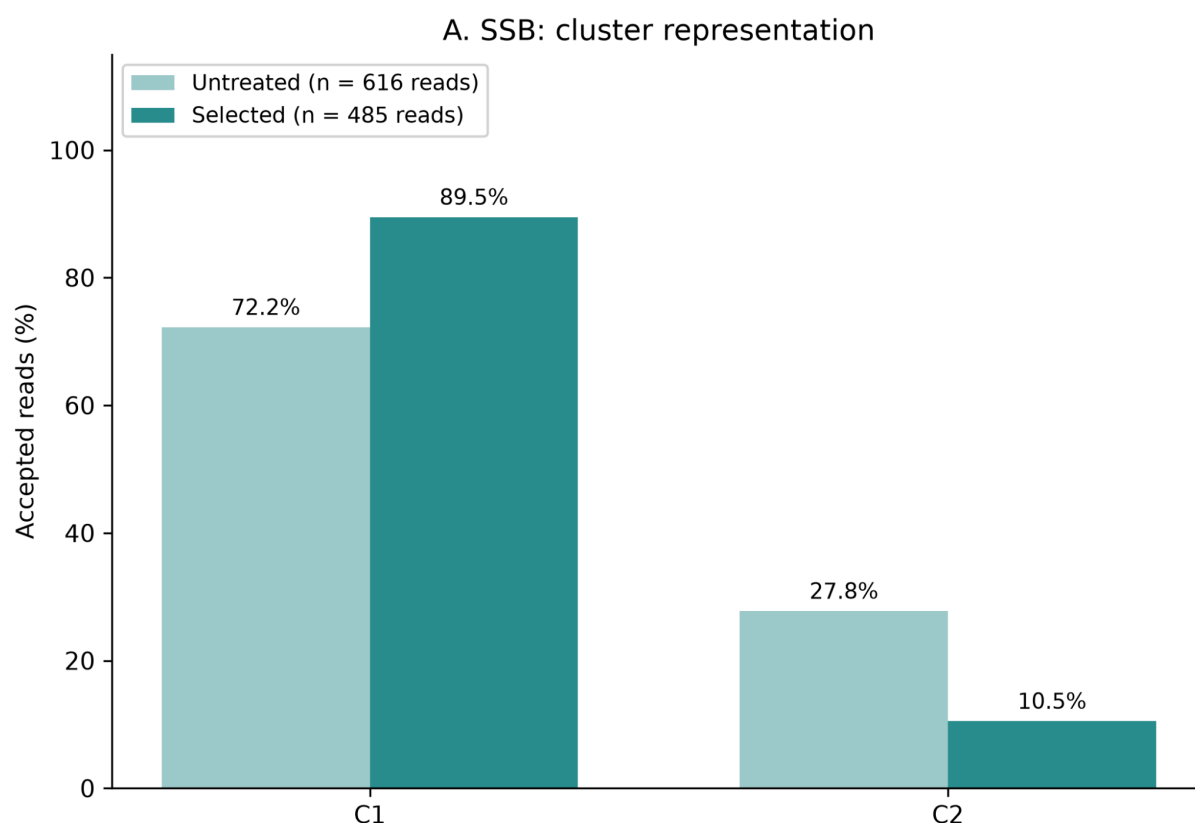


Fig. 11. Cluster representation of SSB clusters before and after selection.

References

1. Cline, J. *et al.* (1996) 'PCR fidelity of Pfu DNA polymerase and other thermostable DNA polymerases,' *Nucleic Acids Research*, 24(18), pp. 3546–3551. <https://doi.org/10.1093/nar/24.18.3546>.
2. Green, M.R. and Sambrook, J. (2018) 'Optimizing primer and probe concentrations for use in Real-Time Polymerase Chain Reaction (PCR) assays,' *Cold Spring Harbor Protocols*, 2018(10), p. pdb.prot095018. <https://doi.org/10.1101/pdb.prot095018>.
3. Jo, B.H. (2022) 'An Intrinsically Disordered Peptide Tag that Confers an Unusual Solubility to Aggregation-Prone Proteins,' *Applied and Environmental Microbiology*, 88(7), p. e0009722. <https://doi.org/10.1128/aem.00097-22>.
4. Le, Y. *et al.* (2013) 'Thermostable DNA Ligase-Mediated PCR Production of Circular Plasmid (PPCP) and Its Application in Directed Evolution via In situ Error-Prone PCR,' *DNA Research*, 20(4), pp. 375–382. <https://doi.org/10.1093/dnares/dst016>.
5. Marblestone, J.G. *et al.* (2005) 'Comparison of SUMO fusion technology with traditional gene fusion systems: Enhanced expression and solubility with SUMO,' *Protein Science*, 15(1), pp. 182–189. <https://doi.org/10.1110/ps.051812706>.

6. Nguyen, T.K.O. *et al.* (2024) 'Efficient production of human interleukin-3 from *Escherichia coli* using protein disulfide isomerase b'a' domain,' *Biotechnology Journal*, 19(5), p. e2300581. <https://doi.org/10.1002/biot.202300581>.
7. Tang, N.C. *et al.* (2024) 'Synthetic intrinsically disordered protein fusion tags that enhance protein solubility,' *Nature Communications*, 15(1), p. 3727. <https://doi.org/10.1038/s41467-024-47519-7>.
8. Yu, X. *et al.* (2016) 'Assessment of the Fusion Tags on Increasing Soluble Production of the Active TEV Protease Variant and Other Target Proteins in *E. coli*,' *Applied Biochemistry and Biotechnology*, 182(2), pp. 769–781. <https://doi.org/10.1007/s12010-016-2360-9>.
9. Zhou, P. and Wagner, G. (2009) 'Overcoming the solubility limit with solubility-enhancement tags: successful applications in biomolecular NMR studies,' *Journal of Biomolecular NMR*, 46(1), pp. 23–31. <https://doi.org/10.1007/s10858-009-9371-6>.